

First two (2) letters of *last* name:

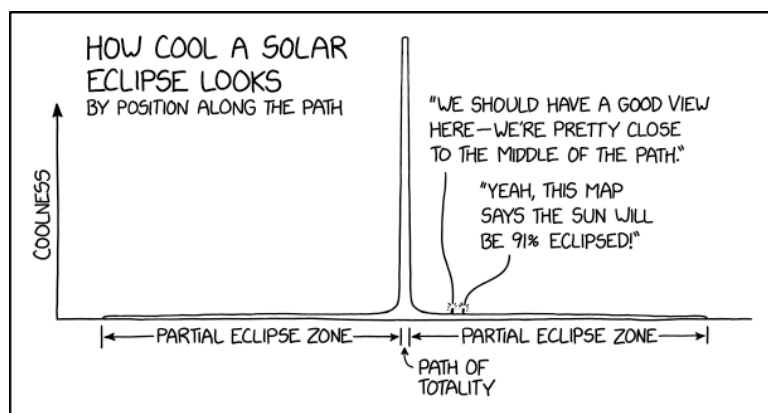
Name (printed legibly): _____

Signature: _____

Student ID #: _____

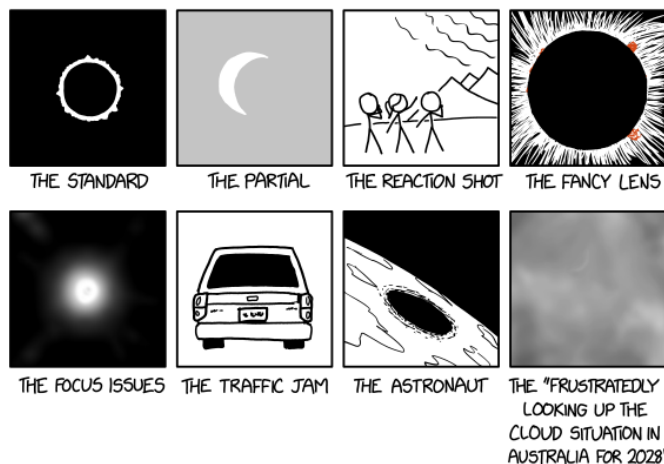
DO NOT OPEN this exam until instructed and the entire class begins.*Opening this prior is a violation of the Academic Integrity Code**as is continuing to write after time is called.*

You may tear the periodic table/scratch paper off the back while waiting to begin.



You have just about completed organic chemistry! Work completely. And work concisely **using drawings!**

xkcd.com/2914 (top) and xkcd.com/2917 (bottom)

TYPES OF ECLIPSE PHOTO

page 2 _____ (19)

page 3 _____ (14)

page 4-5 _____ (36)

page 6 _____ (16)

page 7 _____ (10)

page 8 _____ (6)

TOTAL _____ (100)

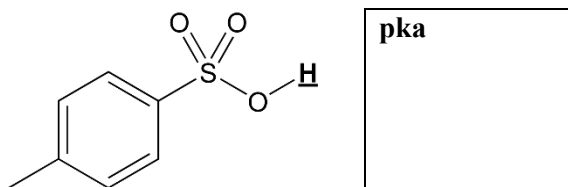


1. Draw a compound containing the following functional groups and no others. (6)

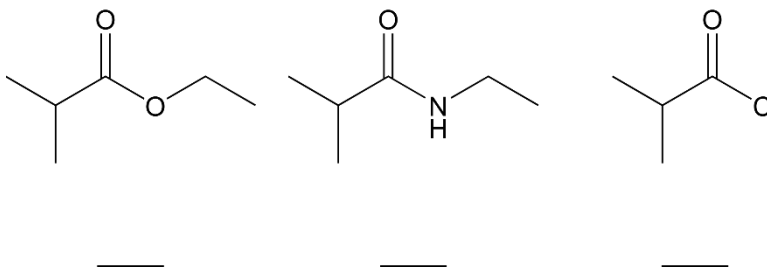
nitrile

anhydride

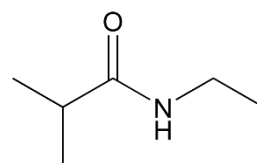
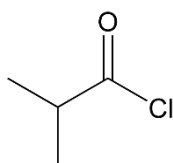
2. Give the pKa of the bold, underlined hydrogen in the following compound. (4)



3. a) Rank the following compounds in order of reactivity towards nucleophiles. 1 = most reactive, 3 = least (3)

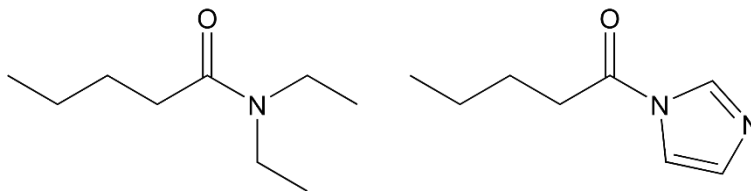


- b) Now use drawings and a phrase to explain the reactivity of the two below. The reason for each will be different. These would support your choices above. (3 points each; 2 points for the drawing, 1 point explanation) (6)



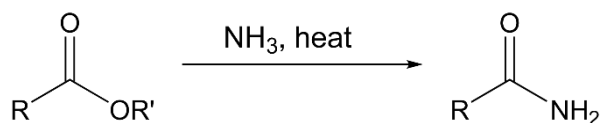


c) Imidazolides (right) are much more susceptible to nucleophilic addition than other amides (e.g. left). Draw a picture (6) illustrating the difference and give a *brief* explanation (3) – this can almost be done in one word. (9)



-
4. Amides can be synthesized by the addition of ammonia to a carboxylic acid or an ester with heating. No acid catalyst is used. With alcohols, a catalysts *must* be used. We discussed two reasons for this difference. (5)

Only 5 points, these can be very short.



a) Why is no acid catalyst needed with amines?

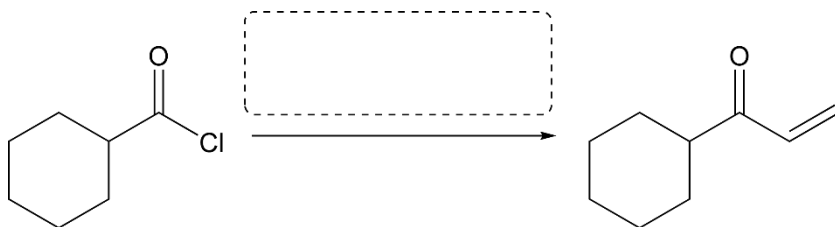
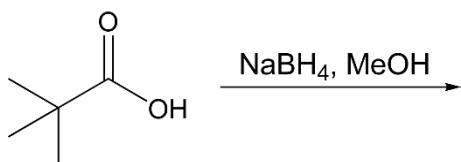
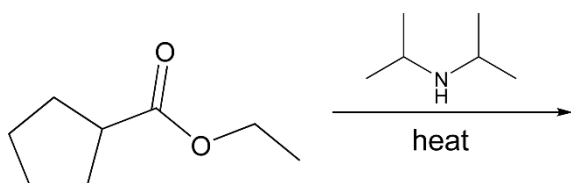
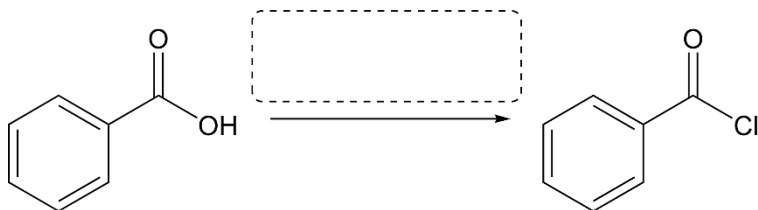
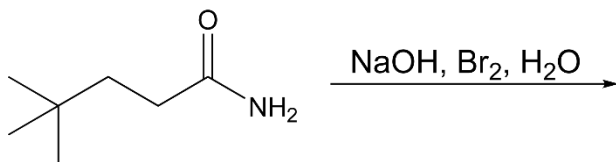
b) What problem arises with the use of acid catalysts with amines?

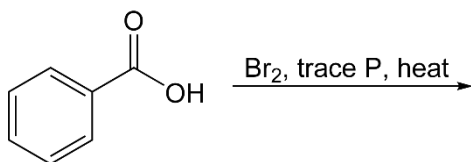
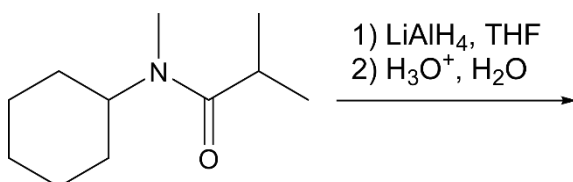
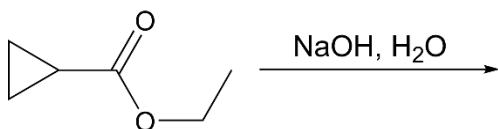
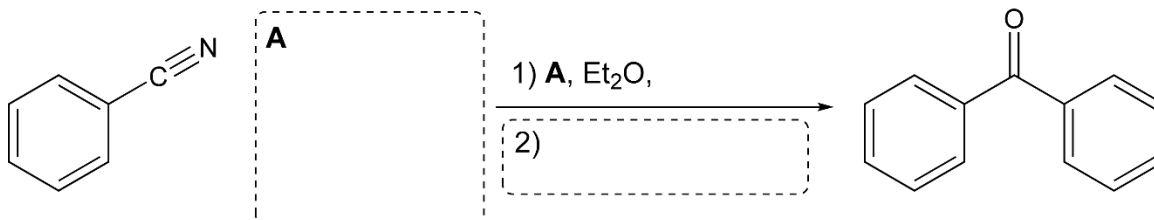
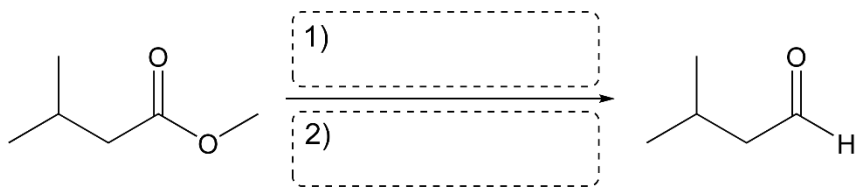


5. Complete the following reactions. In dashed boxes over arrows of reactions where products are shown, fill in the necessary conditions. Give the major product(s) of the other reactions. Be sure to show or otherwise indicate relevant stereochemical outcomes where appropriate. If no reaction takes place, write "no reaction" or "NR".

Write out all reagents. For example, if the Bell Reagent was NaOEt, please write NaOEt or NaOCH₂CH₃ – not "Bell Reagent." (36)

There are 10 reactions. **Complete nine. Write OMIT** in the answer space of the one you do not want graded. You can lose five (5) points if you do not write OMIT.



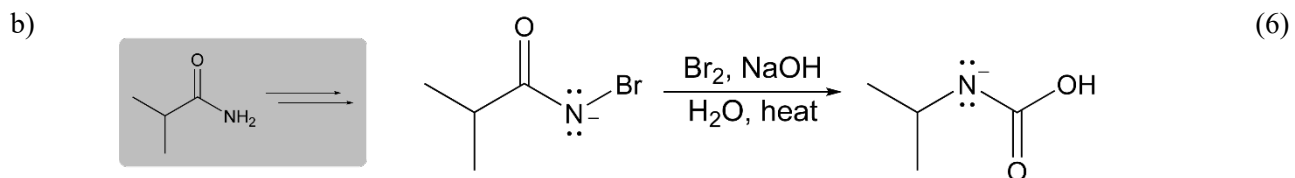
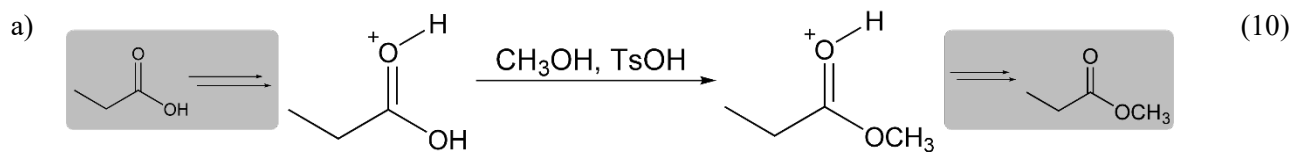


Did you remember to write OMIT for one reaction (in the answer space for that reaction)?



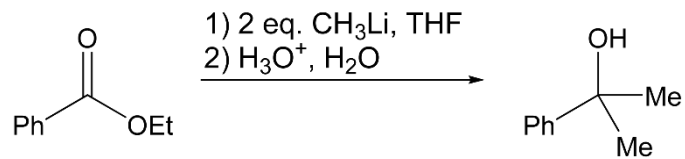
6. Complete the following mechanisms. You only need to give the mechanism for the unshaded reaction. The shaded boxes show the reactants or products for the full reaction.

Write out the complete, step-by-step mechanism using curved arrows for the following reactions. Be certain that there is no ambiguity where arrows begin or end and that all bonds formed or broken are explicitly shown.





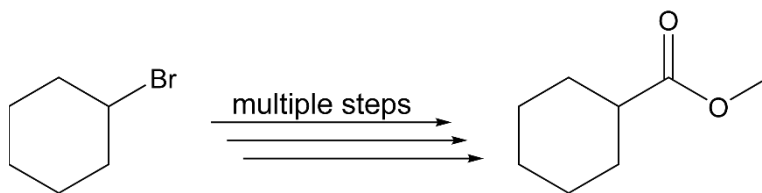
c)



(10)



7. Propose a reasonable synthetic sequence for the following. Just show each individual step and the product of each. You should not draw the mechanisms (use scratch paper to draw those as needed, but just give each reaction here. (6)



Periodic Table of the Elements																																			
1 IA HHydrogen1.008		2 IIA BeBeryllium9.012												13 III A B Boron10.811		14 IV A C Carbon12.011		15 V A N Nitrogen14.007		16 VI A O Oxygen15.999		17 VII A F Fluorine18.998		18 VIII A HeHelium4.003											
3 LiLithium6.941		4 BeBeryllium9.012												5 B Boron10.811		6 C Carbon12.011		7 N Nitrogen14.007		8 O Oxygen15.999		9 F Fluorine18.998		10 Ne Neon20.180											
11 NaSodium22.990		12 MgMagnesium24.305		3 IIB 3 B		4 IVB 4 B		5 VB 5 B		6 VIB 6 B		7 VII B 7 B		8 VIII 8		9 VIII 8		10 X 		11 IB 1 B		12 IIB 2 B		13 Aluminum26.982		14 Silicon28.086		15 P Phosphorus30.974		16 S Sulfur32.066		17 Cl Chlorine35.453		18 Ar Argon39.948	
19 KPotassium39.098		20 CaCalcium40.078		21 ScScandium44.956		22 TiTitanium47.867		23 VVanadium50.942		24 CrChromium51.996		25 MnManganese54.938		26 FeIron55.845		27 CoCobalt58.933		28 NiNickel58.693		29 CuCopper63.546		30 ZnZinc65.38		31 GaGallium69.723		32 GeGermanium72.631		33 AsArsenic74.922		34 SeSelenium78.971		35 BrBromine79.904		36 KrKrypton83.798	
37 RbRubidium85.468		38 SrStrontium87.62		39 Yttrium88.906		40 ZrZirconium91.224		41 NbNiobium92.906		42 MoMolybdenum95.95		43 TcTechnetium98.907		44 RuRuthenium101.07		45 RhRhodium102.906		46 PdPalladium106.42		47 AgSilver107.868		48 CdCadmium112.414		49 InIndium114.818		50 SnTin118.711		51 SbAntimony121.760		52 TeTellurium127.6		53 I Iodine126.904		54 XeXenon131.294	
55 CsCesium132.905		56 BaBarium137.328		57-71		72 HfHafnium178.49		73 TaTantalum180.948		74 WTungsten183.84		75 ReRhenium186.207		76 OsOsmium190.23		77 IrIridium192.217		78 PtPlatinum195.085		79 AuGold196.967		80 HgMercury200.592		81 TlThallium204.383		82 PbLead207.2		83 BiBismuth208.980		84 PoPolonium[208.982]		85 AtAstatine209.987		86 RnRadon222.018	
87 FrFrancium223.020		88 RaRadium226.025		89-103		104 RfRutherfordium[261]		105 DdBDubnium[262]		106 SgSeaborgium[266]		107 BhBohrium[264]		108 HsHassium[269]		109 MtMeitnerium[278]		110 DsDarmstadtium[281]		111 RgRoentgenium[280]		112 CnCoperniciunm[285]		113 NhNihonium[286]		114 FlFlerovium[289]		115 McMoscovium[289]		116 LvLivermorium[293]		117 TsTennessine[294]		118 OgOganesson[294]	